INCH-POUND

MIL-DTL-22520/41
16 June 2008

DETAIL SPECIFICATION SHEET

CRIMPING TOOLS, TERMINAL, HAND, WIRE TERMINATION; TYPE II FOR TERMINALS AND SPLICES SIZE 10-14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-22520.

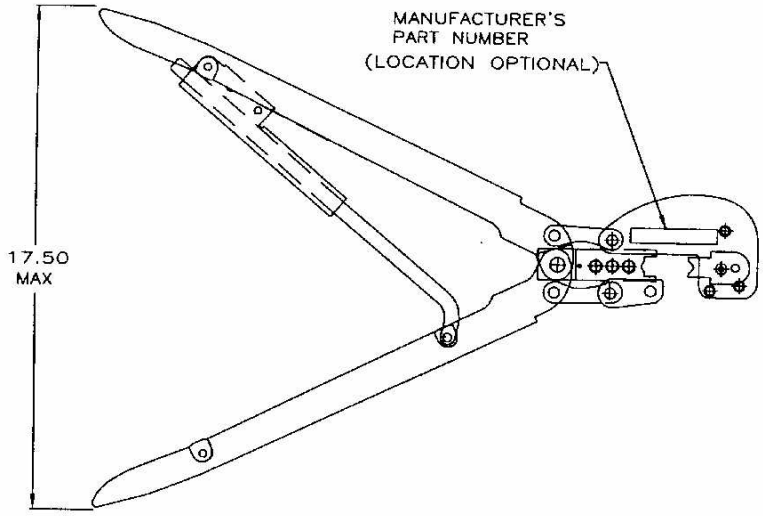


FIGURE 1. Crimp tool configuration.

AMSC N/A FSC 5120

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:50Z
Check the source to verify that this is the current version before use.

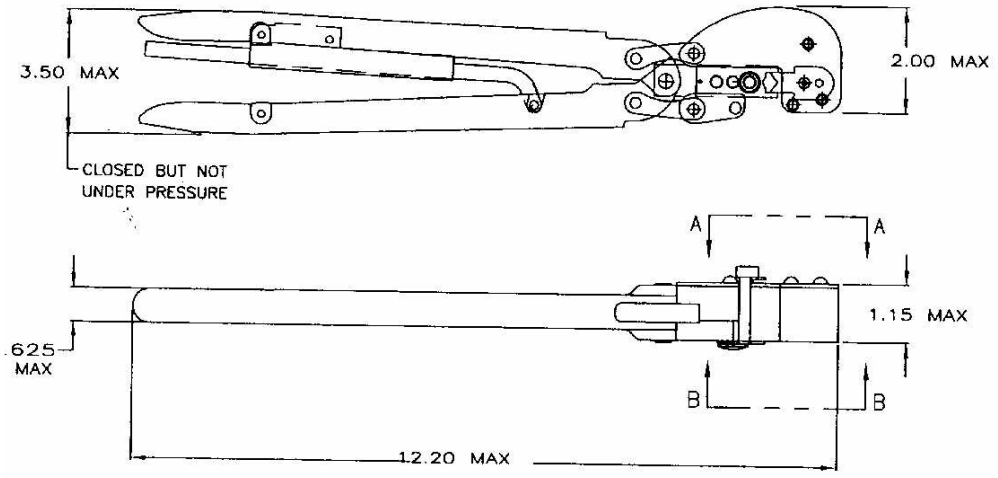


FIGURE 1. Crimp tool configuration - Continued.

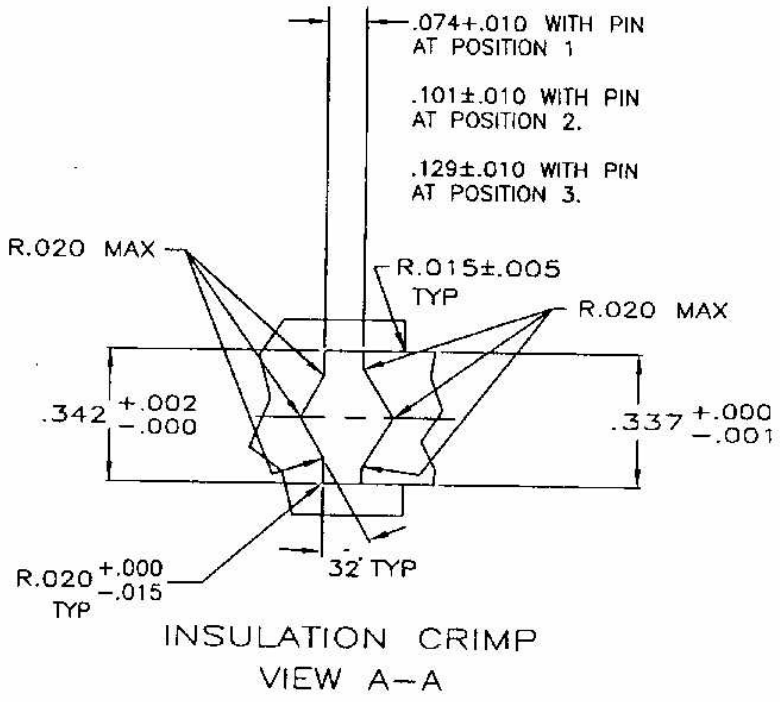


FIGURE 2. Wire barrel crimp dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:50Z
Check the source to verify that this is the current version before use.

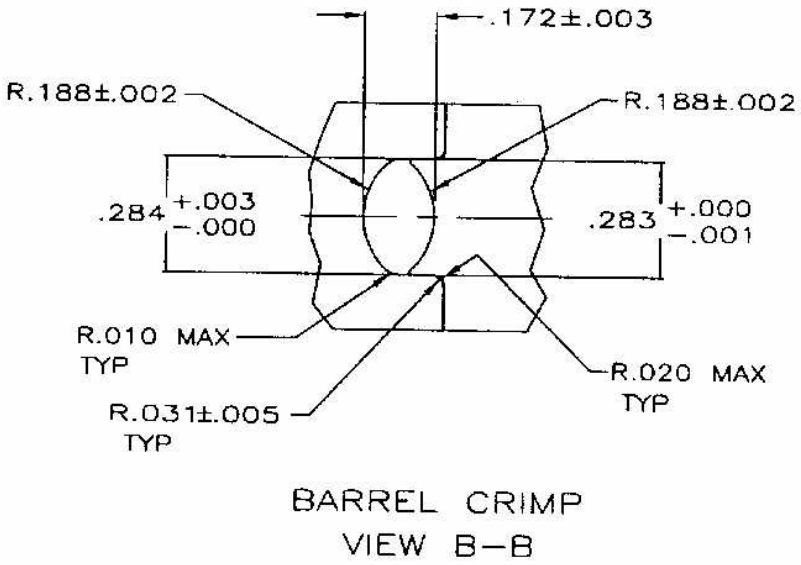


FIGURE 3. Insulation barrel crimp dimensions.

TABLE I. Available gages for tool.

Gage part number	Gage type	Gage area
MIL-DTL-22520/42-01	Go/No-Go	Wire
MIL-DTL-22520/42-02	Go/No-Go	Insulation (position #1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:50Z
Check the source to verify that this is the current version before use.

TABLE II. Metric equivalents.

Inches	MM
0.001	0.025
0.002	0.051
0.003	0.076
0.005	0.127
0.008	0.203
0.010	0.254

Inches	MM
0.015	0.381
0.020	0.508
0.031	0.787
0.074	1.880
0.101	2.565
0.127	3.226
0.129	3.277
0.172	4.369
0.188	4.775

Inches	MM
0.283	7.188
0.284	7.214
0.337	8.560
0.341	8.661
0.342	8.687
0.625	15.875
0.90	22.860
1.15	29.210

Inches	MM
2.00	50.800
3.25	82.550
3.50	88.900
12.20	309.880
17.00	431.800
17.50	449.500

NOTES:

1. Dimensions are in inches.
2. Metric equivalents when specified are based upon 1.00 inch = 25.4 millimeters.
3. Unless otherwise specified herein or in MIL-DTL-22520, tolerances are ± .010 inch (0.25 millimeters) for three place decimals and ± .5° for angles.
4. Refer to MIL-DTL-22520/42 for calibration, inspection gage (M22520/42-01 and -02).

REQUIREMENTS:

1. The envelope dimensions of the tool shall be in accordance with figure 1. Outline configuration optional within maximum dimensions shown for closed handle position.
2. The wire barrel crimp gaging dimensions and insulation barrel crimp gaging dimensions shall be in accordance with figures 2 and 4 when the tool is in the closed handle position.
3. The tool shall not be greater than 2.25 lb (1.02 kg).
4. Material: Jaw - 38 point carbon min. alloy steel Heat Treat Rockwell C 50 – 55 or equivalent.
   Handles - Steel or equivalent.
5. Finish: Black oxide or equivalent.
6. Cycle control (ratchet) adjustment device shall be set to release between 75 to 85 pounds and shall be tamper proof.
7. Convex crimp surfaces shall be polished to 48 (RMS) Microfinish minimum.
8. Each tool shall have an instruction sheet which includes applications, usage and instructions.
9. Tool part number: M22520/41-01
10. Complete manufacturer's name or AIR 1351 listed logo shall be permanently marked on the tool.
11. The tool shall have three (3) adjustment settings to regulate insulation crimp height.
12. Qualification testing shall be in accordance with MIL-DTL-22520 type 2 tool requirements. Initial qualification test samples shall consist of a sample lot of terminals and/or splices from all

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:50Z
Check the source to verify that this is the current version before use.

sources (brands) qualified at the time of tool testing, and the tool is required to meet the minimum requirements with all samples tested.
13. Locator (positioner) design optional providing it meets the performance requirements of all terminals/splices used with the tool.
14. Optional design shall have the same marking requirements as shown on figure 3.
15. No loose parts or components that can become Foreign Object Damage (FOD) hazards are permitted.

APPLICATION:

The crimp cavities are for the crimping of SAE AS25036, SAE AS17143, terminal and splices in sizes 12 - 10.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Army - CR
AF - 11

Preparing activity:
Navy - AS
(Project 5120-2007-002)

Review activities:
Navy - EC, MC, OS, SH
DLA - CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:50Z
Check the source to verify that this is the current version before use.